OAI CU/DU Lab

Deployment and Operation Guide

A tutorial for reproducing the validated configurations

Operator note. This manual describes how to assemble and operate configurations that have worked in the lab. The two USRP B210 radios and the Quectel modem are shared resources. Moving them between hosts is a normal preparation step, not a failed validation.

Version: 29 July 2026

Canonical repository: `https://github.com/promaaa/oai-cu-du-lab/`

GitHub wiki: `https://promaaa.github.io/oai-cu-du-lab/`

Contents

1 Purpose and correct interpretation

This document is an operating tutorial and technical reference reconciled with repository commit `b4652e386d40d9a3c9e1824a325faae5417d6f1d`. It is not a chronological account of unfinished work. The public evidence record and maintained operator matrix document successful demonstrations of the following capabilities:

- monolithic OAI 5G SA with a commercial phone;
- OAI CU/DU split over Ethernet;
- SIB8/PWS transported through F1 and observed on the Nothing Phone;
- F1-C and F1-U over a Wi-Fi GRE tunnel;
- F1 over a Quectel 5G packet session and WireGuard;
- access DU operation on MiniPC, Raspberry Pi 5, and Jetson Orin Nano;
- phone registration, PDU session, public internet, and throughput tests;
- OAI nrUE internet access over an NR radio link;
- repeatable launch, preflight, packet-placement, and rollback through the operator console.

Operator note. A recorded working capability is not fresh-run evidence. Every new run must record the machine-side and phone-side gates separately. The current baseline classifications in this guide therefore remain authoritative even when the supported-capability matrix says `WORKING`.

Only two combined-radio explorations did not reach the intended end-to-end result:

1. using one X310 for the fronthaul and backhaul functions together;
2. using one B210 for the fronthaul and backhaul functions together.

The X310 fronthaul and backhaul functions were usable separately. The combined X310 topology failed. Likewise, a B210 works as an access radio and a B210 can be used in a radio-link experiment, but one B210 could not simultaneously own both independent OAI roles.

2 Hardware pool and movement plan

The lab has two USRP B210 radios and one Quectel modem. Before following a tutorial, place the shared devices according to the selected scenario.

Scenario	B210 placement	Quectel placement
Monolithic reference	One B210 on the monolithic gNB host	Not required
Ethernet split	One B210 on the selected access-DU host	Not required
Wi-Fi GRE split	One B210 on the selected access-DU host	Not required
Quectel split	One B210 for the donor cell and one B210 for the access cell	On the access-DU side as the IP backhaul modem
Pi or Jetson DU	Move an available B210 to that host	Move only when testing Quectel backhaul

The configuration is designed to work correctly with either B210, including on the Raspberry Pi and the Jetson, without relying on a specific device serial. The operator console discovers the connected radio and generates the runtime configuration. Discover the devices live before launch; do not rely on an old USB topology.

2.1 Physical move checklist

1. Stop the current scenario through the operator console.
2. Confirm that no `nr-softmodem` process still owns the radio.
3. Disconnect the B210 or Quectel only after the owning process is stopped.
4. Connect the device to the target host.
5. Confirm the B210 appears at USB SuperSpeed after UHD firmware load.
6. Run UHD discovery and probe before starting OAI.
7. Confirm the intended host, route, and interface with the console read-only checks.

```
uhd_find_devices
uhd_usrp_probe --args serial=<USRP_SERIAL>
lsusb -t
```

3 Architecture

3.1 Monolithic reference

The 5GC and gNB run on Firecell. The gNB owns a local B210 and serves the commercial phone directly. This is the simplest reference for core, RF, registration, PDU, internet, throughput, and PWS.

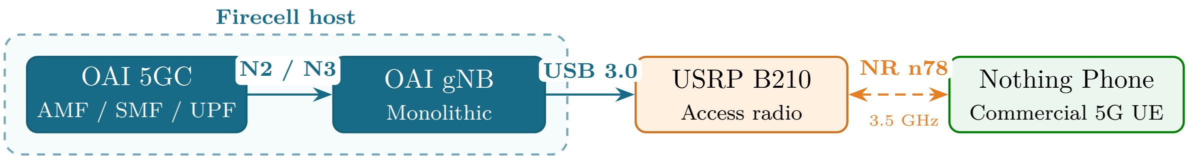

Figure 1: Monolithic OAI 5G SA Reference Architecture

3.2 CU/DU split

Firecell hosts the OAI 5GC and CU. The selected DU host runs MAC, RLC, PHY, and the access B210. F1-C uses SCTP. F1-U uses GTP-U over UDP.

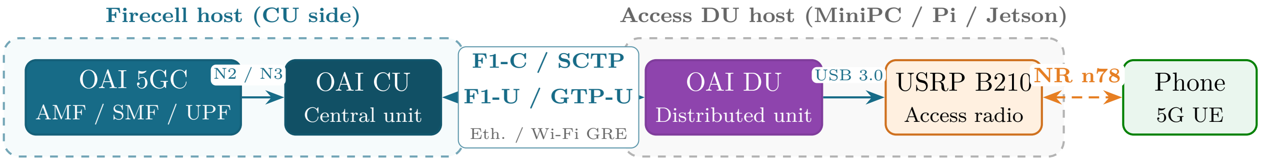

Figure 2: OAI CU/DU Split Architecture

3.3 Quectel wireless F1

The Quectel path replaces the wired F1 transport with a 5G packet session and a WireGuard overlay. The maintained workflow uses a monolithic donor gNB on Firecell only to serve the Quectel modem. This donor has no F1 association. The selected MiniPC, Raspberry Pi, or Jetson access DU is the only DU in this workflow and serves the phone through the second B210.

Warning. The deprecated donor-DU-over-local-F1 experiment is not the maintained Quectel workflow. The modem must not use the access cell as its own backhaul; that creates an unsupported circular dependency.

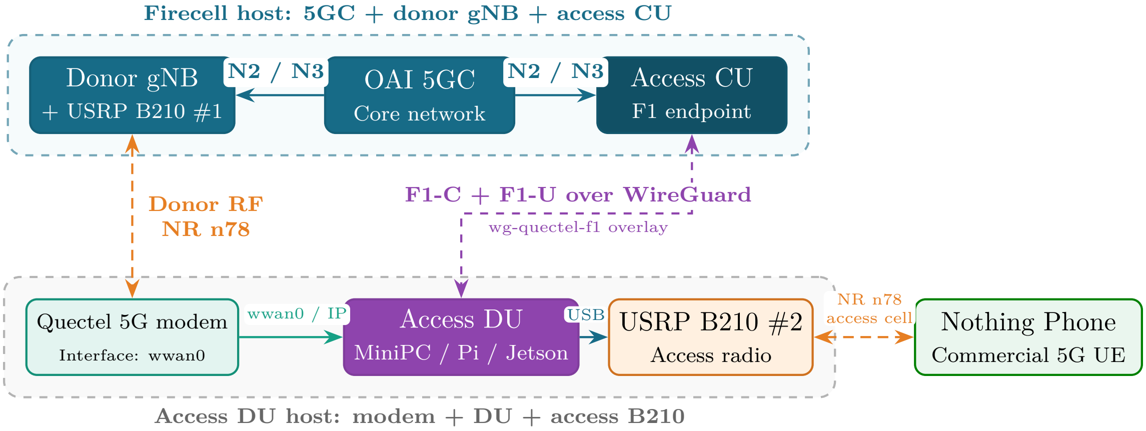

Figure 3: Quectel 5G Wireless Backhaul & WireGuard F1 Architecture

4 Supported capabilities and current baselines

The maintained operator console exposes Ethernet, Wi-Fi GRE, and Quectel with WireGuard for MiniPC, Raspberry Pi, and Jetson access DUs. These nine combinations, the monolithic reference, SIB8 and PWS, phone service, clean stop, and Ethernet rollback are recorded as working capabilities.

Access DU	Ethernet	Wi-Fi GRE	Quectel/WireGuard
MiniPC	`WORKING`	`WORKING`	`WORKING`
Raspberry Pi	`WORKING`	`WORKING`	`WORKING`
Jetson	`WORKING`	`WORKING`	`WORKING`

The table below is the stricter current evidence view. Historical throughput is context, not a ceiling or a substitute for a controlled rerun.

Scenario	Role	Recorded result	Current classification
Monolithic OAI	Reference only	About 150 Mbps historically	`PARTIAL`: machine-side reference reproduced 14 July 2026; phone gates not rerun
MiniPC Ethernet CU/DU + SIB8	Canonical rollback	About 19–23 Mbps historically	`BLOCKED`: B210 was not on MiniPC during the latest rollback attempt
Wi-Fi GRE CU/DU + SIB8	Wireless candidate	About 12 Mbps historically	`HISTORICAL`: requires fresh validation
Raspberry Pi Ethernet CU/DU	Lightweight candidate	About 18–23 Mbps historically	`HISTORICAL`: requires fresh validation
Jetson Ethernet CU/DU	Embedded candidate	6.5–7.3 Mbps in an earlier run	`PARTIAL`: current phone and rollback gates remain incomplete
Quectel/WireGuard F1	Target wireless backhaul	No accepted end-to-end result	`BLOCKED`: the maintained donor/access topology requires fresh proof

Warning. Do not promote a historical throughput number to a current PASS. A full PASS requires all ten validation gates in Section 13 from the same controlled run.

5 Prepare the repository and hosts

5.1 Local configuration

The operator workstation requires Node.js 18 or newer, OpenSSH, lab network access, SSH keys, and a private environment file. Clone the repository and create that ignored local file with private permissions.

```
git clone https://github.com/promaaa/oai-cu-du-lab.git
cd oai-cu-du-lab
mkdir -p conf/local
cp conf/lab.env.example conf/local/lab.env
chmod 600 conf/local/lab.env
```

Edit `conf/local/lab.env` for the current host addresses and runtime paths. Use SSH keys or an agent. Never store passwords, UE authentication values, WireGuard private keys, or subscriber dumps in Git.

The file may instead remain outside the repository:

```
./oai-lab --env="$HOME/private/lab.env" --check-local-setup
```

The maintained split baseline uses external OAI source pinned at:

```
102965a669b9444857c27843ec8ce62780bf9d37
```

Record the repository commit, actual CU and DU OAI commits, and local patch state before a run. Required OAI modifications must be applied from feature-separated patches under `patches/`. Generated runtime configuration remains outside Git.

5.2 Read-only checks

```
./oai-lab --help
./oai-lab --check-local-setup
node scripts/tests/oai-lab-static.test.mjs

./oai-lab --minipc-ethernet --verify
./oai-lab --status
./oai-lab --logs
```

`-check-local-setup` and the static suite do not contact the lab. `-verify`, `-status`, and `-logs` contact the configured hosts. Static tests prove CLI and menu dispatch plus external-command blocking; they are not radio or phone evidence.

Operator note. `./oai-lab` is the only supported operator entry point. Files under `scripts/lib/` are internal implementation details and must not be launched directly.

6 Run the monolithic reference

Use this mode first when checking the core, RF, subscriber, PWS, or user plane. It removes F1 transport from the diagnostic path.

```
./oai-lab --start-mono
./oai-lab --logs
```

1. Connect the Firecell B210, stop any split scenario, and start monolithic mode.
2. Wait for core health, NG setup, RF readiness, and synchronization.
3. Toggle airplane mode once; verify PWS, registration, PDU session, and internet.
4. Record DL/UL throughput, then stop and check for process residue.

The historical reference was about 150 Mbps. The current baseline is `PARTIAL` until the phone gates are rerun.

7 Run Ethernet CU/DU

Place either available B210 on the selected DU host. The operator console discovers its serial and injects it into the runtime configuration. Firecell runs the 5GC and CU; Ethernet carries F1-C and F1-U.

```
./oai-lab --minipc-ethernet --start-ethernet
./oai-lab --pi-ethernet --start-ethernet
./oai-lab --jetson-ethernet --start-ethernet
```

Run only the command for the chosen host, then verify:

1. F1 setup, SCTP, and GTP-U use the selected Ethernet path.
2. The B210 reaches RF-ready and synchronized state.
3. The phone receives PWS, registers, obtains a PDU session, and reaches the internet.
4. DL/UL throughput and clean stop are recorded.

Historical DL context is about 19–23 Mbps on MiniPC, about 18–23 Mbps on Pi, and 6.5–7.3 Mbps on Jetson. None of those numbers replaces a fresh result. On Pi, check the performance governor and throttling. On Jetson, use the SCTP kernel, maximum-performance mode, CPU affinity, and separated USB IRQs.

8 Run Wi-Fi GRE F1

This profile keeps the F1 endpoint addresses but carries F1 over the Wi-Fi underlay and `test-gre` tunnel.

```
./oai-lab --minipc-wifi-gre --start-wifi-gre
./oai-lab --pi-wifi-gre --start-wifi-gre
./oai-lab --jetson-wifi-gre --start-wifi-gre
```

Run only the selected host profile. Before launch, preserve management access and confirm symmetric Wi-Fi/GRE routing. During the run, verify SCTP and GTP-U on `test-gre`, absence of F1 on the direct Ethernet path, and the full phone acceptance sequence from the validation protocol. The current evidence classification is `HISTORICAL`; the roughly 12 Mbps record requires a fresh controlled rerun.

9 Run Quectel/WireGuard F1

This profile requires two independent cells: the monolithic Firecell donor gNB serves the Quectel modem, while the selected access DU and second B210 serve the phone. F1 crosses `wg-quectel-f1` over the detected Quectel data interface, normally `wwan0`; donor and access cell identities must remain distinct.

```
./oai-lab --minipc-quectel-wg --start-quectel-wg
```

```
./oai-lab --pi-quectel-wg --start-quectel-wg
./oai-lab --jetson-quectel-wg --start-quectel-wg
```

Run only the selected host profile, then verify:

1. the donor cell is operational and the modem registers on it;
2. `wwan0` has a packet session and WireGuard tunnel ping succeeds;
3. F1-C/F1-U are inside WireGuard and encrypted outer traffic is on `wwan0`;
4. the phone passes PWS, registration, PDU, internet, and throughput gates.

The current baseline records no accepted end-to-end result for this maintained topology. Treat machine-side WireGuard or F1 success as partial evidence only; do not claim a full PASS until the modem donor-cell identity, access-cell phone traffic, PWS, registration, PDU, internet, throughput, clean stop, and Ethernet rollback are all captured in one controlled run.

10 Send and validate PWS/SIB8

Use `./oai-lab`, select `Change PWS warning text for all TUI configs`, and enter only sanitized warning content. Restart the selected scenario because OAI reads the warning file at startup. The CU sends the warning through F1AP, the DU schedules SIB8, and the phone displays the alert.

Acceptance is simple: confirm the phone is on the intended access cell and visually observe the warning. CU/DU log messages alone are not a PASS.

11 Performance tuning

11.1 BLER and MCS

The operator console generates per-run configuration from the selected host, transport, live B210 serial, and validated defaults. The current default radio window and TCP clamp are:

```
dl_bler_target_lower = 0.45
dl_bler_target_upper = 0.65
ul_bler_target_lower = 0.25
ul_bler_target_upper = 0.45
tcp_mss_clamp        = 1200
```

Ethernet launch first tests whether the full path can carry MTU 9000 before raising both F1 interfaces. If that test fails, the console keeps jumbo frames disabled and uses TCP MSS clamping. Non-Ethernet modes skip jumbo frames.

The default MCS range is asymmetric: DL 10–18 and UL 5–28 for MiniPC and Pi; Jetson uses DL 10–28 and UL 5–28. The `-force-mcs` option raises the minimums to DL 12 and UL 8. Use these maintained defaults first, then change them only from fresh RF, BLER, HARQ, overflow, and throughput evidence.

11.2 Jetson USB and CPU placement

The Jetson requires a clean USB3 path. Keep the DU away from CPU 0 and place the active xHCI interrupt on CPU 0. Select the interrupt that is actually increasing, not simply the first USB-looking line in `/proc/interrupts`.

```
DU process: CPUs 1-5
active xHCI IRQ: CPU 0
```

```
usbfs_memory_mb: 1000
power mode: MAXN_SUPER
```

12 The two unsuccessful combined-radio explorations

12.1 X310 used for fronthaul and backhaul together

The X310 was explored in both roles. The fronthaul function and backhaul function could be run separately at a feasible configuration. The attempt to combine both roles into one end-to-end fronthaul/backhaul system did not work.

The strongest measured transport limitation appeared at 106 PRB. The host to X310 path negotiated only 1 GbE. Both the reduced 46.08 MSps mode and native 61.44 MSps mode overflowed after RF-ready. A new cable rated far above 1 Gb/s did not change the negotiated end-to-end path, so it did not fix the problem.

Table 2: X310 exploration summary

Test	Result
X310 fronthaul role alone	Functional at a feasible standalone profile
X310 backhaul role alone	Functional as a separate role
Combined fronthaul and backhaul	Failed to reach the intended end-to-end result
106 PRB over negotiated 1 GbE	UHD overflow and RFNoC timeout
Higher-rated cable	Did not raise the negotiated path or remove overflow

The next meaningful combined X310 attempt must first prove a real high-speed NIC, switch, MTU, and UHD streaming path. Cable labeling alone is not proof.

12.2 One B210 used for fronthaul and backhaul together

The single-B210 experiment tried to use the two RF chains for independent access and backhaul systems. Two independent OAI processes could not safely open the same UHD device, and both chains shared device-level clock and local oscillator constraints. A single-owner broker moved IQ blocks without queue drops, but the complete donor PRACH, registration, active F1, and phone service chain did not close.

This does not invalidate B210 access operation or B210 radio-link operation. It shows that one B210 cannot be treated as two independent radios by two OAI roles without deeper device-layer integration.

13 Validation protocol

For every tutorial, record these gates separately:

Gate	Required evidence
Host and process health	Correct hosts, commits, processes, core containers, and exclusive ownership
NG and F1-C	NG setup and SCTP association on the selected path
F1-U	GTP-U sockets and packets on the selected path
RF and timing	UHD device, RF ready, sync, and bounded overflow behavior
PWS	Warning visible on the commercial phone
Registration	UE visible as registered in the active core
PDU session	Live data UE address and session rules
External-DN	Ping to the live UE address with recorded loss and RTT
Phone internet	Web or application traffic through the 5G bearer
Throughput	Timestamped DL and UL result with units
Stop and rollback	No stale process, route, firewall, MTU, GRE, or WireGuard state

Runtime evidence is stored outside the repository:

```
~/.local/state/oai-cu-du-lab/runs/
```

Set `OAI_LAB_STATE_DIR` to choose another private location. Commit only sanitized conclusions.

14 Recovery and troubleshooting

Symptom	First action
PWS but no 5G service	Watch RACH and initial UL RRC while toggling airplane mode once
UE registered but no PDU	Inspect AMF, SMF, NRF, and DNN state before restarting only the failing service
PDU exists but no phone internet	Confirm live UE IP, external-DN ping, APN, NAT, and UPF route
B210 at 480M	Probe with UHD and verify runtime re-enumeration at 5000M
Jetson overflow or loss	Check active xHCI IRQ placement and the overflow delta during traffic
X310 106 PRB overflow	Stop and prove negotiated transport above 1 GbE first
F1 on the wrong interface	Stop through the operator console, inspect routes and tunnels, then relaunch

Recovery order:

1. Stop the selected scenario through the operator console.
2. Confirm remaining softmodem and core processes.
3. Remove only scenario-owned route, tunnel, firewall, and MTU changes.
4. Preserve management connectivity.
5. Move the shared hardware to the rollback arrangement.
6. Restore the MiniPC Ethernet profile and validate it end to end.

15 Public evidence provenance

This tutorial relies only on the sanitized evidence published in `docs/evidence/`, the pinned artifact manifest, and the maintained baseline and status documents. Private laboratory notes, raw captures, operational addresses, and intermediate interpretations are intentionally outside the publication corpus.

16 Quick command reference

```
# Interactive operator interface
./oai-lab

# Read-only checks
./oai-lab --help
./oai-lab --check-local-setup
./oai-lab --status
./oai-lab --logs
./oai-lab --minipc-ethernet --verify

# MiniPC profiles
./oai-lab --minipc-ethernet --start-ethernet
./oai-lab --minipc-wifi-gre --start-wifi-gre
./oai-lab --minipc-quectel-wg --start-quectel-wg

# Raspberry Pi profiles
./oai-lab --pi-ethernet --start-ethernet
./oai-lab --pi-wifi-gre --start-wifi-gre
./oai-lab --pi-quectel-wg --start-quectel-wg

# Jetson profiles
./oai-lab --jetson-ethernet --start-ethernet
./oai-lab --jetson-wifi-gre --start-wifi-gre
./oai-lab --jetson-quectel-wg --start-quectel-wg

# Local static audit, no lab contact
node scripts/tests/oai-lab-static.test.mjs
```

17 Conclusion

The project records a functioning multi-configuration OAI platform. Monolithic, Ethernet split, Wi-Fi GRE, Quectel/WireGuard, Pi DU, Jetson DU, PWS, phone user plane, and software UE internet have all worked. The shared hardware must be moved to assemble the selected configuration, which is normal for this lab. These capability records do not replace the current baseline classifications or the need for fresh evidence.

The two negative results are narrow and useful: one X310 did not complete the combined fronthaul/backhaul topology even though the roles worked separately, and one B210 did not complete simultaneous fronthaul and backhaul. These results define hardware and transport boundaries; they do not redefine the successful configurations as failures.